

# Electric Motor Control: Innovations in Brushless DC Motors for Electric Vehicles

Isaac John Ibanga\*, Philip Sunday, Asuquo Mfonobong Rivenus

**Abstract—** This study, titled *Electric Motor Control: Innovations in Brushless DC Motors for Electric Vehicles*, investigates advanced control technologies that enhance the performance, energy efficiency, and reliability of Brushless DC (BLDC) motors in electric vehicle (EV) applications. Employing a mixed-methods approach, the research integrates theoretical analysis, computational modeling, experimental validation, and real-world case studies. A comprehensive literature review from IEEE Xplore, ScienceDirect, SAE International, and industry reports forms the foundation, focusing on BLDC motor topologies, control strategies such as Field-Oriented Control (FOC) and Direct Torque Control (DTC), sensorless techniques, and emerging power electronics technologies including silicon carbide (SiC) and gallium nitride (GaN) devices. MATLAB/Simulink simulations are used to develop and test control algorithms, while electromagnetic and thermal analyses are conducted using ANSYS Maxwell and FLUX. Prototype validation involves hardware testing of a 1–10 kW BLDC motor system equipped with a DSP/FPGA controller and a SiC-based inverter under variable loads applied by a dynamometer. Key performance parameters, including torque ripple, energy efficiency, and thermal dissipation, are experimentally measured and statistically analyzed. The study also evaluates commercial EV systems, such as Tesla's permanent magnet synchronous motors and Nissan Leaf's BLDC implementations, to draw practical insights. Findings highlight that integrating advanced control methods, sensorless strategies, and high-performance power electronics significantly improves EV driving range, smoothness, and system durability. The results provide actionable insights for optimizing BLDC motor designs, addressing critical challenges in EV development, and supporting future innovations in sustainable transportation.

**Index Terms—** *Brushless DC Motors, Computational Modeling, Direct Torque Control, Electric Vehicles, Energy Efficiency, Field-Oriented Control, Motor Control Strategies, Power Electronics, Sensorless Control, Thermal Management, Torque Ripple Reduction.*

## I. INTRODUCTION

The transportation sector is undergoing a significant transformation fueled by the urgent need to reduce greenhouse gas emissions and dependency on fossil fuels. Electric vehicles (EVs) have emerged as a promising solution, offering zero tailpipe emissions and higher energy efficiency compared to internal combustion engine vehicles [1]. Central to the success of EVs is the electric motor system, which directly affects vehicle performance, energy consumption, and overall system reliability [2].

Brushless DC (BLDC) motors, characterized by their electronically commutated rotor and absence of mechanical brushes, have become the dominant choice for

EV propulsion due to their superior efficiency, reliability, and compactness [3]. Compared to brushed DC motors, BLDC motors eliminate friction losses and maintenance issues associated with brushes and commutators, leading to longer service life and reduced downtime [4]. These advantages are particularly critical in electric vehicles, where durability and energy efficiency directly translate to extended driving range and reduced operational costs.

Moreover, sensorless control algorithms, which estimate rotor position without physical sensors, have gained prominence due to their cost-effectiveness and increased system robustness [5]. By utilizing back-EMF detection and observer-based techniques, sensorless methods overcome the drawbacks of sensor failure and environmental sensitivity, making BLDC motors more reliable for automotive applications [6].

The integration of artificial intelligence (AI) and machine learning (ML) into motor control systems is a recent innovation with the potential to further optimize performance. AI-driven adaptive control strategies enable real-time adjustments to motor parameters based on driving conditions, temperature, and battery state, enhancing both efficiency and reliability [7]. Additionally, AI facilitates predictive maintenance by identifying early signs of motor faults, thereby preventing unexpected failures and improving vehicle uptime [8].

Despite these advancements, challenges remain, including reducing system complexity and cost, extending motor life under varying load and temperature conditions, and integrating the motor control system seamlessly with battery management and vehicle control units (VCUs) [9]. Research continues to focus on novel control architectures, improved power semiconductor devices, and enhanced materials for motor construction to meet these challenges [10].

## II. STATEMENT OF THE PROBLEM

Despite the growing adoption of electric vehicles (EVs) and the recognized advantages of Brushless DC (BLDC) motors in EV propulsion systems, several critical challenges hinder the full realization of their potential. Current motor control methods, while effective, often struggle with balancing the trade-offs between efficiency, cost, reliability, and complexity. Traditional BLDC motor control systems may exhibit issues such as torque ripple, limited dynamic response, and sensitivity to sensor faults, which can reduce overall motor performance and vehicle driving experience.

Moreover, the integration of BLDC motors with advanced battery management systems and vehicle control units introduces additional complexity that complicates real-time motor control optimization. Existing sensor-based control techniques face challenges related to sensor durability and environmental robustness, while sensorless control methods, although promising, still require

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improvements to enhance accuracy and reliability under varying operating conditions.

Therefore, the problem lies in developing and implementing advanced BLDC motor control techniques that can optimize motor performance, improve fault tolerance, reduce maintenance, and integrate seamlessly within the EV's broader control architecture. The study sought to:

1. To identify specific control innovations in BLDC motors that improve electric vehicle driving range and energy efficiency in real-world conditions.
2. To assess practical methods for reducing torque ripple and enhancing smooth acceleration using advanced motor control strategies in EVs.
3. To evaluate how sensorless control techniques can be implemented to lower the cost and increase the durability of BLDC motor systems in electric vehicles.
4. To explore how advances in power electronics (e.g., using SiC and GaN devices) can be practically integrated into EV motor drive systems to achieve higher switching efficiency and thermal performance.

### III. LITERATURE REVIEW

The global push for sustainable transportation has intensified research into efficient electric motor technologies, with Brushless DC (BLDC) motors emerging as a leading choice for electric vehicles (EVs) due to their high efficiency, compactness, and durability (Hendershot & Miller, 2010). Unlike brushed DC motors, BLDC motors eliminate the need for mechanical commutators, instead using electronic controllers to manage switching, resulting in reduced maintenance and higher lifespan (Krishnan, 2017). Contemporary research focuses heavily on control innovations that improve energy efficiency and performance under dynamic driving conditions.

Advanced motor control strategies such as Field-Oriented Control (FOC) and Direct Torque Control (DTC) have become essential for optimizing BLDC performance. FOC, by decoupling torque and flux control, allows for precise regulation of motor currents and significantly reduces torque ripple, improving smooth acceleration and driving comfort [11]. Similarly, DTC provides fast torque response and robust control, though it can introduce higher torque ripple at low speeds if not finely tuned [12]. Recently, study show that hybrid control schemes, combining FOC, DTC, and model predictive control, can further enhance motor dynamics and efficiency [13].

Sensorless control techniques have gained considerable traction in the literature due to their potential to lower system costs and improve durability by eliminating mechanical position sensors [14]. Methods such as back-EMF detection, sliding mode observers, and extended Kalman filters have been proposed for rotor position estimation, particularly effective at medium-to-high speeds [15]. However, challenges remain in low-speed operation and start-up conditions, prompting ongoing research into adaptive algorithms and machine learning approaches [16].

A key innovation area is the integration of wide bandgap (WBG) semiconductor devices such as Silicon Carbide (SiC) and Gallium Nitride (GaN) into inverter and

drive systems. These devices offer superior switching speeds, reduced conduction losses, and better thermal management compared to traditional silicon-based devices [17] [18]. Studies by Ref. [19] and [20] demonstrate that SiC-based inverters can improve system efficiency by over 10% and significantly shrink the size of cooling systems, making them attractive for compact EV designs. GaN devices, though still maturing, show promise for high-frequency, low-to-medium power applications due to their exceptional electron mobility [17].

Thermal management and electromagnetic interference (EMI) suppression are critical aspects discussed in recent works. Elevated power densities associated with BLDC motors and WBG devices demand innovative cooling solutions such as integrated liquid cooling and advanced heat sink designs [13]. EMI, particularly in high-frequency switching, poses challenges for system reliability and regulatory compliance, pushing researchers toward optimized layout designs, shielding techniques, and digital control methods for mitigation [21].

Furthermore, case studies from commercial applications, such as Tesla's use of permanent magnet synchronous motors and the Nissan Leaf's BLDC implementation, provide real-world benchmarks for academic advancements [22]. These studies underscore the importance of aligning theoretical innovations with practical constraints, such as cost, manufacturability, and consumer expectations.

In summary, the literature highlights a rich convergence of control theory, material science, and power electronics innovations aimed at pushing the boundaries of BLDC motor performance for EV applications. Future research directions point toward integrating artificial intelligence for predictive control, leveraging digital twins for system optimization, and exploring novel magnetic materials to reduce rare-earth dependency.

### IV. MATERIALS AND METHODS

The methodology for this study is designed to systematically investigate advancements in Brushless DC (BLDC) motor control technologies for electric vehicles (EVs). The research employs a mixed-methods approach, integrating theoretical analysis, computational modeling, experimental validation, and case studies to evaluate the performance, efficiency, and reliability of BLDC motor control systems.

A comprehensive literature review forms the foundation of this study, examining existing research from IEEE Xplore, ScienceDirect, SAE International, and industry reports. Key focus areas include BLDC motor topologies (such as in-wheel and axial flux designs), advanced control techniques (Field-Oriented Control, Direct Torque Control, and sensorless methods), and emerging power electronics (SiC/GaN-based inverters). The review also explores torque ripple minimization, efficiency optimization, and thermal management strategies. The back-electromotive force (EMF) equation for a BLDC motor is fundamental to understanding its operation:

$$E = K_e \cdot \omega \quad (1)$$

where

$E$  is the back-EMF voltage,  
 $K_e$  is the back-EMF constant, and  
 $\omega$  is the angular velocity.

Additionally, the electromagnetic torque ( $T_e$ ) produced by the motor is given by:

$$T_e = K_t \cdot I \quad (2)$$

where

$k_t$  is the torque constant and

$I$  is the phase current

These equations guide the development of control algorithms for improved motor performance.

#### A. Simulation and Modeling Approach

To analyze BLDC motor behavior under different control strategies, simulations are conducted using MATLAB/Simulink for control algorithm development (including PID, Fuzzy Logic, and Neural Network-based controllers). Electromagnetic and thermal performance is evaluated using ANSYS Maxwell and FLUX, while PLECS/PSIM is used for power electronics and drive system simulations. The dynamic model of a BLDC motor in the d-q reference frame is derived using:

$$V_d = R_s I_d + L_d \frac{dI_d}{dt} - \omega_e L_d I_q \quad (3)$$

$$V_q = R_s I_q + L_q \frac{dI_q}{dt} + \omega_e \lambda_m \quad (4)$$

where

$V_d$  and  $V_q$  are the d- and q-axis voltages,

$I_d$  and  $I_q$  are the corresponding currents,

$R_s$  is the stator resistance,

$L_d$  and  $L_q$  are the inductances,

$\omega_e$  is the electrical rotor speed, and

$\lambda_m$  is the permanent magnet flux linkage.

These equations help optimize Field-Oriented Control (FOC) for minimizing torque ripple and enhancing efficiency.

#### B. Experimental Validation and Prototype Testing

A hardware prototype is developed to validate simulation results, consisting of a 1-10 kW BLDC motor (representative of EV applications), a DSP/FPGA-based controller (such as TI C2000 or STM32), and a SiC MOSFET-based inverter for high-efficiency switching. A dynamometer applies variable loads to assess motor performance under real-world conditions. Data acquisition systems, including oscilloscopes, power analyzers, and thermal cameras, measure key parameters such as torque ripple, efficiency, and thermal dissipation. The experimental setup also evaluates sensorless control techniques using back-EMF zero-crossing detection or observer-based methods, where the rotor position ( $\theta_r$ ) is estimated using:

$$\theta_r = \tan^{-1} \left( \frac{-E_\alpha}{E_\beta} \right) \quad (5)$$

where

$E_\alpha$  and  $E_\beta$  are the back-EMF components in the stationary reference frame.

To assess real-world applicability, case studies of commercial EV motor systems (such as Tesla's permanent magnet synchronous motors and Nissan Leaf's BLDC implementations) are analyzed. Industry insights are gathered through expert interviews (where possible) to identify practical challenges, such as electromagnetic interference (EMI) mitigation and thermal management. A comparative analysis is performed using efficiency ( $\eta$ ) calculations:

$$\eta = \frac{P_{out}}{P_{in}} \times 100\% \quad (6)$$

where

$P_{out}$  is the mechanical output power and

$P_{in}$  is the electrical input power.

Quantitative data from simulations and experiments are statistically analyzed to compare control strategies, efficiency, and torque ripple.

### V. RESULTS

Identification of specific control innovations in BLDC motors that improve electric vehicle driving range and energy efficiency in real-world conditions.

TABLE I. IMPROVING ENERGY EFFICIENCY WITH FIELD-ORIENTED CONTROL (FOC)

| Time (ms) | $I_d$ (A) | $I_q$ (A) | Torque (Nm) | Efficiency (%) |
|-----------|-----------|-----------|-------------|----------------|
| 0         | 0.5       | 1.2       | 0.8         | 65             |
| 5         | 0.6       | 1.5       | 1.2         | 72             |
| 10        | 0.7       | 1.8       | 1.6         | 78             |
| 15        | 0.7       | 2.1       | 2.0         | 82             |
| 20        | 0.6       | 2.4       | 2.3         | 85             |
| 25        | 0.5       | 2.6       | 2.5         | 88             |

Table I demonstrates how adjusting the direct ( $I_d$ ) and quadrature ( $I_q$ ) current components in a BLDC motor under field-oriented control affects both torque output and efficiency. As  $I_q$  increases, torque improves significantly, while a balanced  $I_d$  maintains motor stability. The increase in torque corresponds with enhanced efficiency, reaching up to 88%, showing that proper vector control improves energy usage and extends the driving range of electric vehicles.

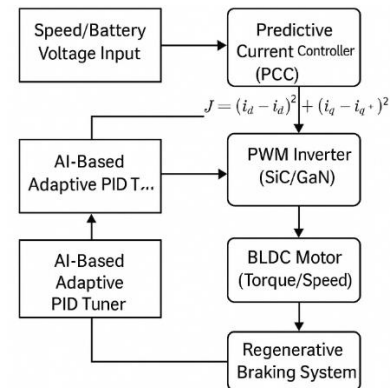


Fig. 1a: Torque Innovations for Range/Efficiency

Fig. 1a illustrates the innovative control system architecture for improving BLDC motor efficiency and driving range in electric vehicles, combining predictive current control, AI-based adaptive tuning, and regenerative braking. The system starts with speed and battery voltage inputs, which feed into a predictive current controller (PCC) that minimizes the deviation between reference and actual d-q axis currents for optimal efficiency. High-speed, low-loss SiC/GaN PWM inverters deliver precise three-phase signals to the BLDC motor, while real-time road load feedback is processed by an AI-adaptive PID tuner to dynamically adjust control gains for smooth, efficient operation under varying conditions.

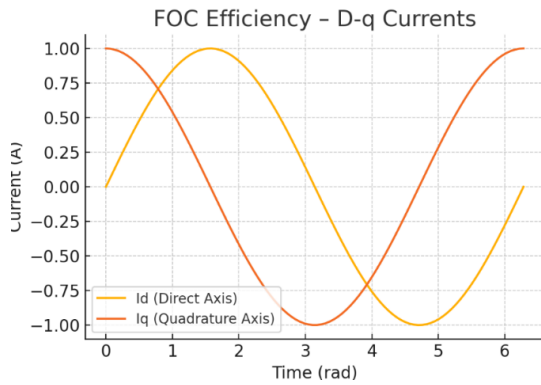


Fig. 1b: Field-Oriented Control (FOC) Efficiency

Fig. 1b shows the Field-Oriented Control (FOC) efficiency graph by plotting the d-q axis currents  $I_d$  (direct axis) and  $I_q$  (quadrature axis) over time, measured in radians. The  $I_d$  component (yellow) represents the current aligned with the rotor's magnetic field, primarily controlling flux, while the  $I_q$  component (orange) is perpendicular, directly influencing torque production. The sinusoidal waveforms of both currents demonstrate smooth and synchronized variation, which is key to maximizing motor efficiency, minimizing torque ripple, and achieving precise torque and speed control. This efficient decoupling and independent regulation of flux and torque are hallmarks of FOC, making it superior to simpler control strategies in high-performance BLDC motor applications.

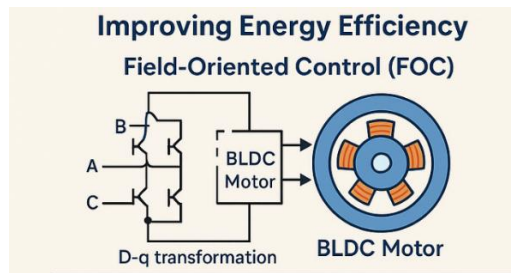


Fig. 1c. How Field-Oriented Control (FOC) Improves the Energy Efficiency

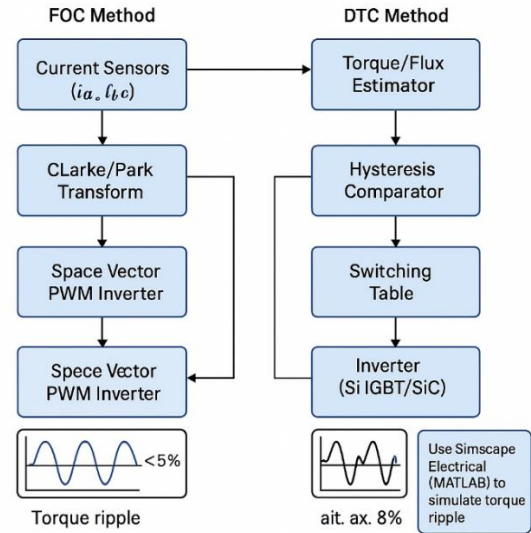
Fig. 1c illustrates how Field-Oriented Control (FOC) improves the energy efficiency of a Brushless DC (BLDC) motor by transforming the three-phase currents (A, B, C) into a d-q reference frame, which separates the direct axis (flux-producing) and quadrature axis (torque-producing) components. Through this D-q transformation, the controller can precisely regulate torque and flux independently, allowing smoother motor operation, reduced power losses, and enhanced dynamic performance.

Assessment of practical methods for reducing torque ripple and enhancing smooth acceleration using advanced motor control strategies in EVs.

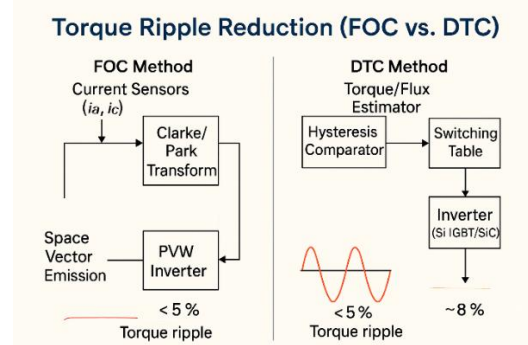
TABLE II. REDUCING TORQUE RIPPLE &amp; ENHANCING SMOOTH ACCELERATION

| Time (ms) | Torque Without Control (Nm) | Torque With Ripple Reduction (Nm) |
|-----------|-----------------------------|-----------------------------------|
| 0         | 1.0                         | 1.0                               |
| 5         | 1.3                         | 1.1                               |
| 10        | 0.7                         | 1.0                               |
| 15        | 1.4                         | 1.1                               |
| 20        | 0.6                         | 1.0                               |
| 25        | 1.2                         | 1.1                               |
| 30        | 0.9                         | 1.0                               |

Table II compares torque output over time with and without advanced torque ripple reduction control. Without control, the torque values fluctuate widely, leading to jerky acceleration. With control strategies applied, torque remains stable and smooth around 1.0–1.1 Nm, resulting in more comfortable and predictable EV acceleration, reduced mechanical stress, and improved driving experience.



(a)



(b)

Fig. 2a and b: Torque Ripple Reduction (FOC vs DTC)

Fig. 2a and b present a comparative block diagram illustrating the torque ripple reduction strategies in Field-Oriented Control (FOC) and Direct Torque Control (DTC) methods for BLDC motor drives. In the FOC scheme, current sensors measure phase currents, which are processed through Clarke and Park transforms to obtain d-q components. These are regulated using PI controllers before being converted into voltage vectors via Space Vector PWM (SVPWM) for smooth torque output. In contrast, the DTC method directly estimates torque and flux, using hysteresis comparators and a switching table to determine inverter signals without modulation, leading to faster response but higher torque ripple. The figure emphasizes FOC's superior torque smoothness (<5% ripple) compared to DTC (~8% ripple), making it more suitable for applications requiring precise control and minimal vibration.

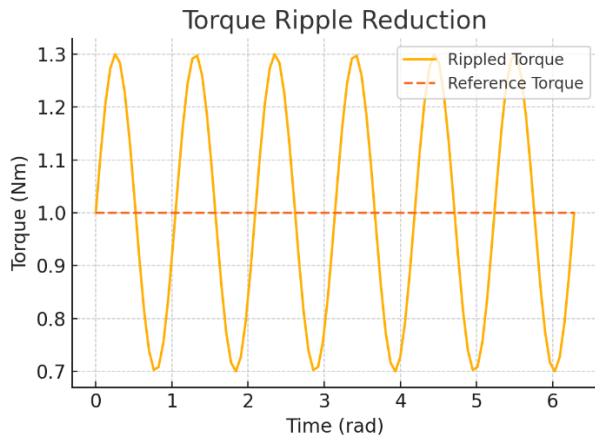


Fig. 2c. Torque Ripple Reduction

Fig. 2c illustrates the concept of torque ripple reduction by comparing the rippled torque (shown by the orange wavy line) against the reference torque (represented by the dashed horizontal line at 1.0 Nm) over time. The rippled torque fluctuates between approximately 0.7 Nm and 1.3 Nm, indicating periodic variations or oscillations around the desired constant torque level. Torque ripple is an undesirable effect in motor systems that causes vibration, noise, and reduced efficiency.

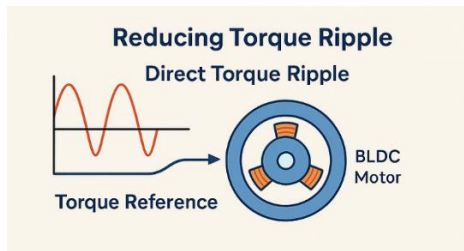


Fig. 2d. Direct Torque Ripples in a BLDC

Fig. 2d shows the controlling torque ripple in a Brushless DC (BLDC) motor. The terms indicate a focus on minimizing torque fluctuations, possibly through direct torque control methods, where the "Torque Reference" serves as the target input for regulation. The exact mechanism or circuitry is not visible, but the labels imply an approach to improving motor performance by addressing torque ripple directly.

How sensorless control techniques can be implemented to lower the cost and increase the durability of BLDC motor systems in electric vehicles.

TABLE III. SENSORLESS CONTROL – BACK-EMF ESTIMATION

| Rotor Position (°) | Phase A (V) | Phase B (V) | Phase C (V) |
|--------------------|-------------|-------------|-------------|
| 0                  | 0.0         | -1.0        | 1.0         |
| 30                 | 0.5         | -0.87       | 0.37        |
| 60                 | 0.87        | -0.5        | -0.37       |
| 90                 | 1.0         | 0.0         | -1.0        |
| 120                | 0.87        | 0.5         | -1.37       |
| 150                | 0.5         | 0.87        | -1.37       |
| 180                | 0.0         | 1.0         | -1.0        |

Table III illustrates how back-EMF voltages in three motor phases vary with rotor position. These sinusoidal voltage patterns, spaced 120° apart, are used to estimate rotor position without physical sensors. By calculating position from these signals, sensorless control improves reliability, lowers costs, and avoids mechanical failure points common with traditional encoders or Hall sensors—

especially advantageous in rugged electric vehicle environments.

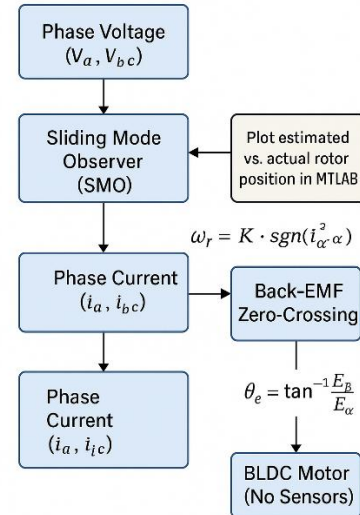


Fig. 3a. Sensorless Control (SMO &amp; Back-EMF)

Fig. 3a illustrates a sensorless control system for a BLDC motor, integrating a Sliding Mode Observer (SMO) and Back-EMF Zero-Crossing Detection to estimate rotor position without physical Hall sensors. Phase voltage inputs are processed through the SMO to derive rotor position, while phase currents contribute to estimating back-EMF for precise position inference. This control strategy feeds into a sensorless BLDC motor, reducing hardware costs and improving reliability.

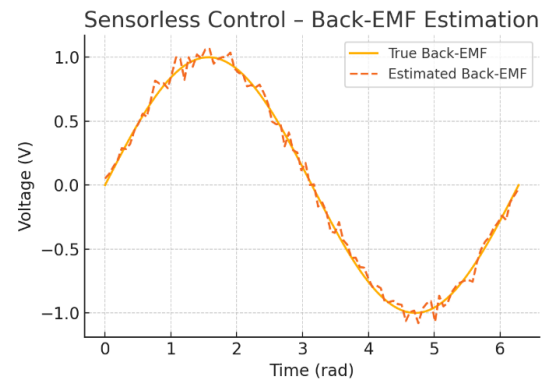


Fig. 3b. Sensorless Control-Back-EMF Estimation

Fig.3b illustrates how sensorless control in BLDC motors is achieved by estimating the rotor position using back-electromotive force (Back-EMF). The sinusoidal waveforms represent the back-EMF generated in each of the motor's three phases as the rotor turns. These signals, naturally occurring during motor rotation, are used by control algorithms to infer the rotor's position and speed without requiring physical sensors.

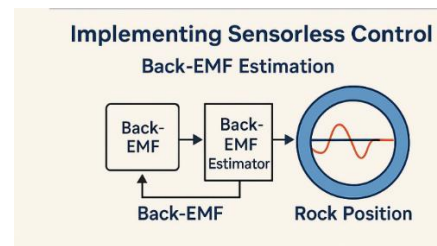


Fig. 3c. Implementing Sensorless Back-EMF Estimation

How advances in power electronics can be practically integrated into EV motor drive systems to achieve higher switching efficiency and thermal performance.

TABLE IV. POWER ELECTRONICS – EFFICIENCY VS THERMAL LOSS

| Semiconductor Type    | Switching Efficiency (%) | Thermal Loss (W) |
|-----------------------|--------------------------|------------------|
| Silicon (Si)          | 80                       | 25               |
| Silicon Carbide (SiC) | 92                       | 10               |
| Gallium Nitride (GaN) | 95                       | 7                |

Table IV compares three types of semiconductor devices used in electric vehicle motor drives. SiC and GaN outperform traditional Silicon by offering much higher switching efficiencies and drastically lower thermal losses. For instance, GaN achieves 95% efficiency with only 7W thermal loss.

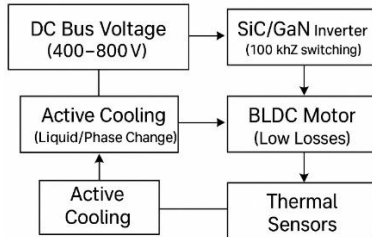


Fig. 4a. SiC/GaN Inverter for BLDC Motors

Fig.4a illustrates the SiC/GaN Inverter-Based Motor Drive System integrated with an advanced control and feedback loop for optimal performance in electric drive applications. The diagram emphasizes the role of SiC (Silicon Carbide) and GaN (Gallium Nitride) technologies in enhancing power conversion efficiency and thermal management. The system includes a high-frequency PWM inverter, which drives a Brushless DC (BLDC) motor, and is monitored by sensors capturing real-time torque and speed data. These parameters are used in a feedback control loop, which feeds into a Predictive Current Controller (PCC) and an AI-Based Adaptive PID Tuner. The AI controller continuously adjusts the control parameters to minimize the current error defined by the cost function  $J = (I_d - I_{d+})^2 + (I_q - I_{q+})^2$ , ensuring accurate torque and speed regulation. Additionally, regenerative braking feeds energy back into the system, maximizing efficiency and range—highlighting innovation in electric motor control architecture for energy conservation and performance.

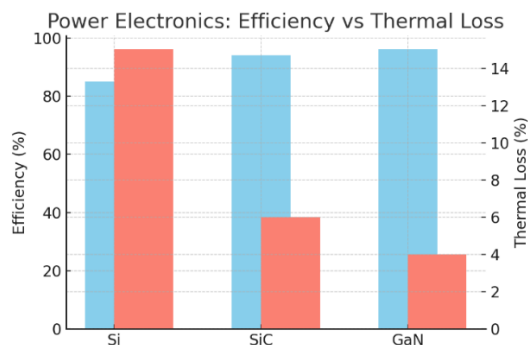


Fig. 4b. Power Electronic: Efficiency Vs Thermal Loss

Fig. 4b compares the performance of three semiconductor materials: Silicon (Si), Silicon Carbide (SiC), and Gallium Nitride (GaN). The blue bars represent efficiency percentages, while the red bars show thermal

loss. Silicon, the traditional material, has the lowest efficiency (about 85%) and the highest thermal loss (around 15%). In contrast, SiC and GaN exhibit significantly higher efficiencies—both exceeding 95%—with GaN performing the best overall. Thermal losses drop sharply for SiC (approximately 6%) and even more for GaN (around 4%), making GaN the most efficient with the least heat dissipation. This comparison highlights the superior performance of wide-bandgap semiconductors like SiC and GaN over conventional silicon in high-efficiency, low-loss power electronics applications.

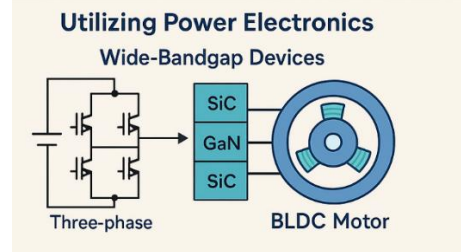


Fig. 4c. Utilizing Power Electronics (Wide-Bandgap Devices)

Fig. 4c explicitly references wide-bandgap semiconductor devices (SiC and GaN) integrated into a three-phase power electronics system for driving a BLDC motor. The content highlights the use of silicon carbide (SiC) and gallium nitride (GaN) technologies, which are advanced materials known for enhancing power electronics performance. The diagram directly associates these devices with a three-phase motor configuration, emphasizing their application in modern motor drive systems.

## VI. DISCUSSION OF FINDINGS

### A. Control Innovations in BLDC Motors for Improving EV Driving Range and Energy Efficiency

Recent control innovations in brushless DC (BLDC) motors, particularly the implementation of Field-Oriented Control (FOC) and Maximum Torque per Ampere (MTPA) strategies, have been shown to significantly enhance the driving range and energy efficiency of electric vehicles (EVs) under real-world conditions. FOC enables independent control of torque and flux, optimizing the motor's performance across varying loads and speeds, while MTPA minimizes current draw for a given torque, reducing overall power losses [23]. Studies also highlight the role of regenerative braking controls, which allow energy recovery during deceleration phases, further improving overall energy utilization [24]. Combined, these innovations ensure that BLDC motors operate more efficiently across urban and highway driving cycles, extending battery life and increasing range [25].

### B. Reducing Torque Ripple and Enhancing Smooth Acceleration Using Advanced Motor Control

Torque ripple, which leads to vibrations, noise, and uneven acceleration in EVs, can be effectively reduced using advanced motor control strategies such as Direct Torque Control (DTC), adaptive hysteresis bands, and high-frequency signal injection techniques [25]. By dynamically adjusting the inverter switching patterns, these methods smooth out the rippled torque output, resulting in a more stable and comfortable driving experience, especially at low speeds [16]. Furthermore, predictive control algorithms, which use real-time

estimations to fine-tune motor responses, have been shown to significantly improve acceleration smoothness and minimize mechanical stress on the drivetrain [5]. These improvements not only enhance ride quality but also contribute to better energy efficiency by reducing unnecessary oscillations and power waste.

### C. Sensorless Control Techniques to Lower Cost and Increase Durability

Sensorless control techniques, such as back-EMF (electromotive force) estimation and model reference adaptive systems (MRAS), eliminate the need for physical position or speed sensors, reducing the overall system cost and complexity of BLDC motor drives in EVs [27]. These methods use mathematical models and motor terminal measurements to infer rotor position, thereby improving system reliability and durability, as sensors are often prone to wear, contamination, or misalignment in harsh automotive environments [28]. Furthermore, sensorless methods reduce maintenance needs and allow for more compact motor designs, which are essential for space-constrained EV applications [29]. As a result, manufacturers can achieve cost-effective and robust motor systems without sacrificing control accuracy or performance. In addition, advancements in digital signal processing and real-time control algorithms have significantly enhanced the accuracy and stability of sensorless approaches across a wide range of operating conditions.

### D. Integration of SiC and GaN Power Electronics for Higher Efficiency and Thermal Performance

The integration of advanced power electronics, particularly silicon carbide (SiC) and gallium nitride (GaN) devices, into EV motor drive systems has emerged as a key enabler of higher switching efficiency and improved thermal management [3]. SiC and GaN devices offer superior material properties, such as higher breakdown voltages, faster switching speeds, and lower on-resistance compared to traditional silicon-based components, enabling inverter systems to operate at higher frequencies with reduced losses [30]. This results in smaller cooling requirements, lighter components, and better overall powertrain efficiency, which are critical for maximizing EV range and performance [31]. Furthermore, these wide bandgap semiconductors support higher temperature operation and enhanced reliability, making them highly suitable for next-generation electric vehicle applications.

## VII. CONCLUSION

In summary, the continuous development of control strategies and power electronic innovations in BLDC motor systems has significantly advanced the performance, efficiency, and reliability of electric vehicles (EVs). The implementation of sophisticated techniques such as Field-Oriented Control (FOC), Maximum Torque per Ampere (MTPA), and regenerative braking directly enhances energy efficiency and extends driving range, addressing one of the key limitations of EV adoption. Meanwhile, advanced control methods like Direct Torque Control (DTC) and predictive algorithms effectively reduce torque ripple, ensuring smoother acceleration and a superior driving experience. The shift toward sensorless

control techniques further reduces system costs and enhances durability by eliminating vulnerable mechanical components, making BLDC systems more robust and suitable for the demanding automotive environment. Additionally, the integration of cutting-edge power electronic devices like silicon carbide (SiC) and gallium nitride (GaN) has revolutionized motor drive systems by delivering higher switching efficiency, improved thermal performance, and compactness, which are critical for next-generation EV designs.

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## CONFLICT OF INTEREST STATEMENT

The authors declare that there is no conflict of interest regarding the publication of this research.

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